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Smart Helmet with Alcohol Detection & Intelligent Safety System

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INTRODUCTION

Road accidents are increasing due to:
Drunk driving
Rash driving
Lack of safety systems
Need for a smart solution to improve rider safety

02.

OUR PROJECTS

Smart Helmet with Alcohol Detection & Intelligent Safety System

03.

ABOUT US

Elaborate on what you want to discuss.

Objective & Problem Statement

Objective

To design a smart helmet that:
Detects accidents
Prevents drunk driving
Sends emergency alerts
Tracks rider location

Problem Statement

Many accidents happen due to:
Alcohol consumption
Delay in emergency response
No immediate system to inform family or ambulance

Proposed Solution

A smart helmet system that:
Detects alcohol before riding
Monitors accidents
Sends real-time alerts
Connects via Bluetooth

OUR PROJECT

Microcontroller Platform: Most prototypes use the Arduino Uno or Arduino Nano as the central brain to process sensor data and control the vehicle's ignition.Sensor Network: It relies on an MQ-3 alcohol sensor for breath analysis and Infrared (IR) sensors to verify helmet compliance.Wireless Communication: Data is transmitted between the helmet and the bike using RF (Radio Frequency) modules or Bluetooth.Emergency Alerting: In case of a crash, it utilizes GPS (NEO-6M) for location tracking and GSM (SIM800L) to send automated emergency SMS

System Features

System Features

GPS Tracking
Emergency Alert System
Bluetooth Connectivity
Alcohol Detection Sensor
Accident Detection

CASE STUDIES

CASE STUDY 1: AI IN HEALTHCARE

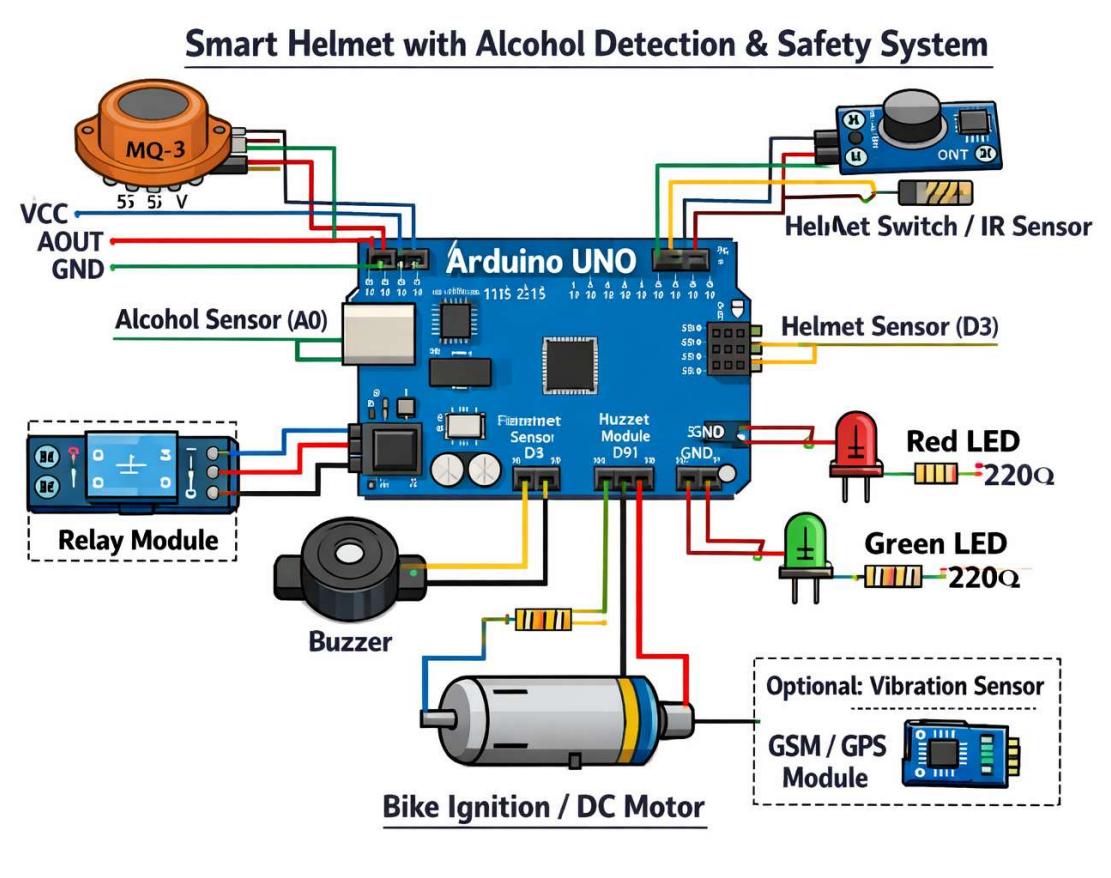
The integration of Artificial Intelligence (AI) into healthcare has brought about a transformative shift in how medical diagnosis, treatment, and patient care are approached. However, while AI offers numerous benefits, its use in these critical areas also raises significant ethical concerns that must be carefully considered to ensure that technology serves the best interests of all patients.

CASE STUDY 2: AI IN AUTONOMOUS VEHICLES

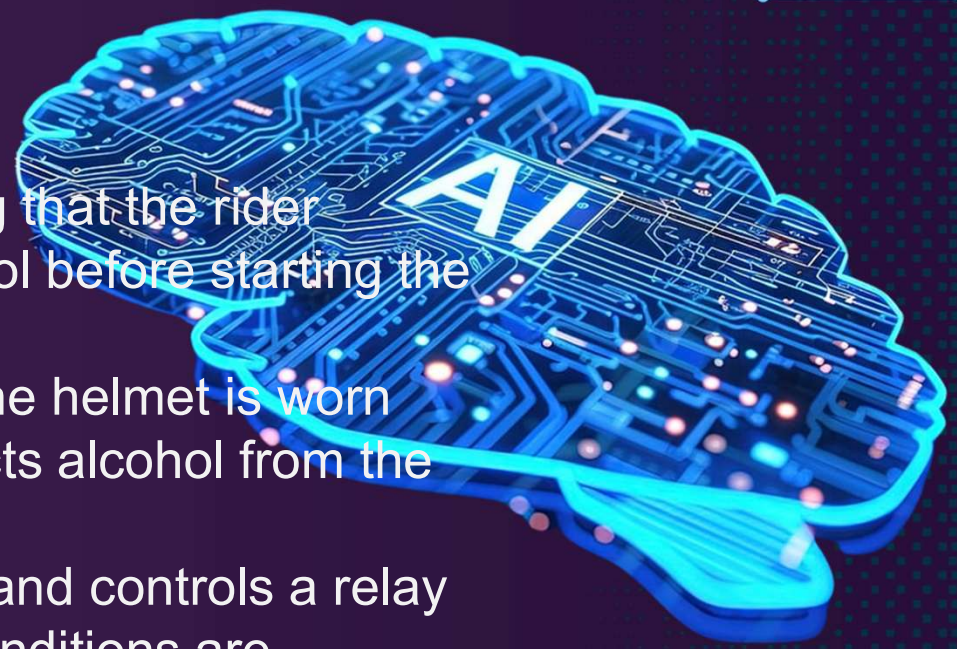
Autonomous vehicles (AVs) represent a groundbreaking innovation in transportation, promising to reduce traffic accidents, improve road safety, and revolutionize mobility. However, the deployment of AVs also introduces profound ethical dilemmas, particularly in scenarios where the vehicle must make decisions involving life-and-death situations. Unlike human drivers.

Block Diagram

Alcohol Sensor → Arduino → Relay (Ignition Control)
MPU6050 → Arduino → GSM + GPS → Alert Message
Bluetooth → Mobile



Working of the smart system



This project improves rider safety by ensuring that the rider wears a helmet and has not consumed alcohol before starting the vehicle.

A helmet detection sensor checks whether the helmet is worn properly, while an MQ-3 alcohol sensor detects alcohol from the rider's breath.

The Arduino processes these sensor inputs and controls a relay connected to the ignition system. If unsafe conditions are detected, the bike ignition remains OFF and an alert is generated using buzzer and LED indicators. In the advanced version, accident detection and emergency SMS alert can also be included

Emergency Alarm System

A. GSM Alert
If accident happens:
Send SMS to family:
“Accident detected! Location

B. GPS Tracking
Share exact location

C. OLED / LCD Display
Display:
Helmet: YES/NO
Alcohol: SAFE/UNSAFE
Bike: START/STOP

D. Bluetooth App
Helmet status on mobile
E. RF Module
Helmet and bike communicate wirelessly

Future Scope & Conclusion

Future Scope

AI-based accident prediction
Mobile app integration
Voice assistant system
Cloud data storage

Conclusion

Smart helmet improves road safety
Combines prevention + detection
Helps save lives